



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT2-Technical Presentation

CSA03 –Data Structures

Presented by:
Shaik Ayesha Tabassum

A database indexing system needs fast insertion, deletion, and searching operations. Recommend an appropriate tree data structure and justify your choice

Recommendation: B-Tree

A **B-Tree** is the most appropriate tree data structure for a database indexing system because it provides **fast insertion, deletion, and searching operations** while efficiently handling large amounts of data stored on disks.

INTRODUCTION

- A database indexing system helps retrieve data quickly and efficiently.
- Large databases require fast searching, insertion, and deletion of records.
- Tree data structures are commonly used to organize indexed data.
- A **B-Tree** is a self-balancing tree that stores multiple keys in each node.
- It reduces disk access and improves database performance.
- B-Trees are widely used in modern Database Management Systems (DBMS).

Why B-Tree?

Fast Searching :

- Search time: **$O(\log n)$**
- Quickly finds records in large databases.

Fast Insertion:

- Inserts new records without making the tree unbalanced.

Fast Deletion:

- Deletes records efficiently using merge and redistribution operations.

Balanced Structure:

- All leaf nodes remain at the same level.
- Performance stays consistent.

Optimized for Disk Storage:

- Stores multiple keys in one node.
- Reduces the number of disk reads and writes.

Objectives

- To understand the role of tree data structures in database indexing.
- To recommend a suitable tree structure for fast searching, insertion, and deletion.
- To explain why **B-Tree** is preferred in database management systems.
- To study the advantages and applications of B-Trees in handling large datasets.
- To improve knowledge of efficient data organization and retrieval techniques.



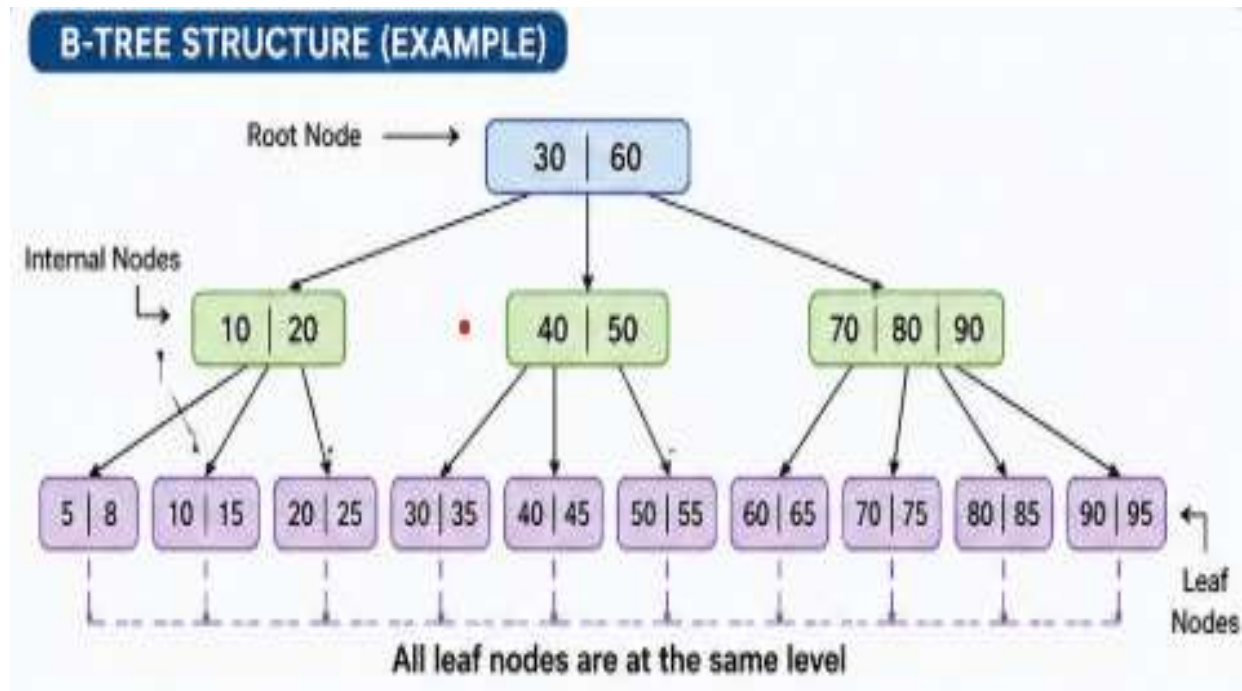
PROBLEM STATEMENT

- Database systems store a large amount of data that must be accessed quickly.
- Efficient searching, insertion, and deletion are essential for good performance.
- Traditional data structures may become slow as the database grows.
- Frequent disk access can reduce system efficiency.
- A suitable tree structure is needed to maintain balanced performance.

Working of B-Tree

- Data is stored in sorted order.
- Search starts from the root node.
- If the node becomes full during insertion, it is split.
- During deletion, nodes are merged or keys are redistributed.
- The tree always remains balanced.

B-Tree Structure:



Advantages

- High search speed
- Efficient insertion and deletion
- Reduces disk I/O operations
- Suitable for large databases
- Self-balancing
- Used in database indexing and file systems

Applications

- Database Management Systems (DBMS)
- File Systems
- Search Engines
- Database Indexing
- Large Data Storage Systems

Conclusion

- **B-Tree** is the most suitable tree data structure for database indexing systems.
- It provides **fast searching, insertion, and deletion** with **$O(\log n)$** time complexity.
- Its balanced structure ensures consistent performance even for large databases.
- It minimizes disk I/O by storing multiple keys in each node.
- Therefore, B-Trees are widely used in **DBMS, file systems, and indexing applications** because they improve data access speed, efficiency, and scalability.

Thank You